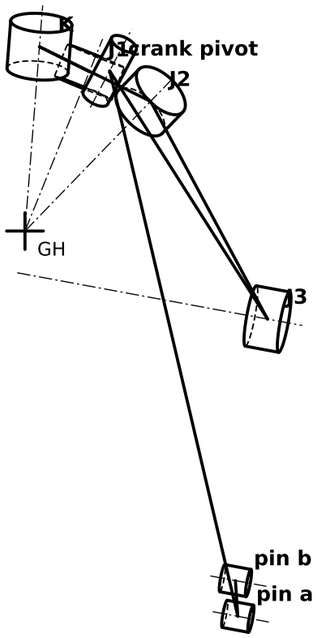
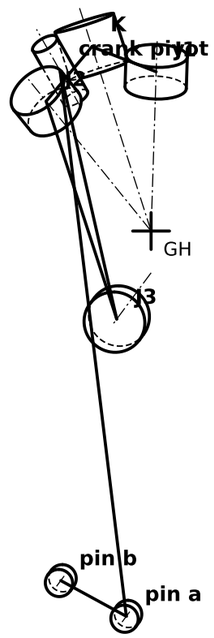
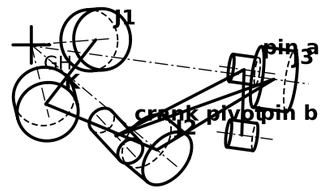
FROM BEHIND (z right, y up)



FROM THE RIGHT SIDE (x right, y up)



FROM ABOVE (z right, x up)



JOINT AXES IN THE BODY FRAME (origin: GH centre; x anterior, y up, z lateral; rest pose; mm)

JOINT	AXIS i, j, k (outward)	ELEV.	AZIM.	REF. POINT x, y, z	FACE FROM GH	AXIS OFF GH	HUB H	∠ TO PREV.	GAP TO PREV.	HUB CTR DIST.
J1	+0.0295 +0.9267 +0.3746	67.93°	85.49°	1.49, 46.77, 18.91	50.47	0.00	11.15			
K	-0.3048 +0.9495 +0.0744	71.71°	166.28°	-16.80, 52.33, 4.10	55.12	0.00	16.35	26.00°	0.00	27.76
J2	-0.5025 +0.6211 +0.6014	38.40°	129.88°	-31.41, 38.83, 37.60	62.52	0.00	11.15	38.03°	0.00	43.05
J3	-0.1339 -0.1802 +0.9745	-10.38°	97.82°	-10.46, -27.69, 73.66	78.17	13.84	11.15	57.22°	7.13	83.81
crank pivot	-0.4487 +0.7900 +0.4178	52.18°	137.05°	-24.66, 43.41, 22.96	54.95	0.00	22.30	71.04°	5.97	97.66
pin a	-0.1339 -0.1802 +0.9745	-10.38°	97.82°	-7.49, -124.82, 65.14	86.97	110.97	8.75	71.04°	44.17	183.94
pin b	-0.1339 -0.1802 +0.9745	-10.38°	97.82°	-28.86, -113.17, 64.35	86.97	101.09	8.75	0.00°	23.19	24.35

REF. POINT: on the axis at the joint's skin-side face (the lower member's face). FACE FROM GH: that face's distance from GH along the axis.
AXIS OFF GH: closest distance from GH to the axis line (0 = the axis passes through the shoulder centre). HUB H: skin-side face to crown.
ELEV.: angle above horizontal. AZIM.: angle in the horizontal plane from anterior (+x) toward lateral (+z).
∠ / GAP TO PREV.: angle and closest distance between this axis and the previous joint's axis (the link between them). HUB CTR DIST.: straight distance between hub mid-points.

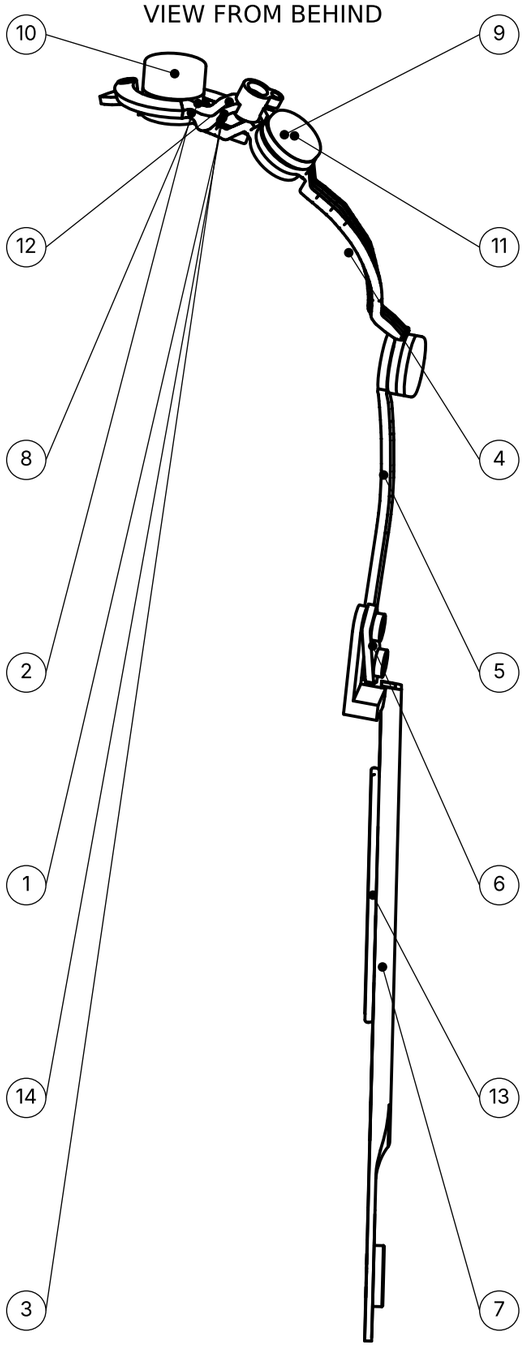
NOTES

1. Use this sheet with the part sheets: each part's END POSITIONS are its slice of this table, in its own frame.
2. J1, K, J2 axes pass through GH (spherical chain); J3 is offset; K is slaved by the cam drive. Knee pins are parallel to J3.
3. Rest pose of the built CAD (arm at the side). Joint angles change with the arm; the part-frame geometry does not.

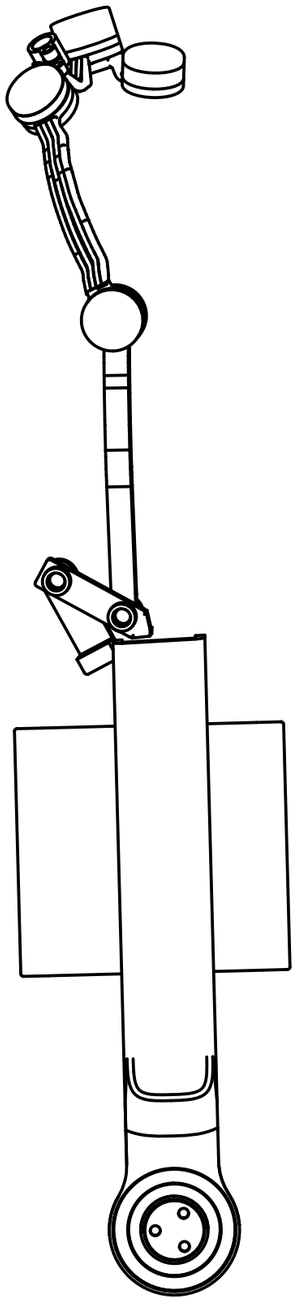
THIRD ANGLE

TITLE: **Cone-cut chain, kinematic layout**		DRAWING NO.: **EXO-A-KIN**
DESIGN: conecut-knee-knuckle-3r (3-joint lead)	PART NO.: **A-KIN**	REV / STATUS: A · for review
MATERIAL: –	MASS: –	QTY PER ASSEMBLY: 1
FINISH: –	SCALE: 1:2	SHEET: 1 / 15
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: joint axes from the part STEP files

VIEW FROM BEHIND



VIEW FROM THE RIGHT SIDE



PARTS LIST

ITEM	PART NO.	DESCRIPTION	QTY
1	D8-01	Scapula base (J1 upper member)	1
2	D8-02a	Link 1a (J1 to K, flank lug)	1
3	D8-02b	Link 1b (K to J2, crank pin)	1
4	D8-03	Link 2 (J2 to J3)	1
5	D8-04	Member 3 (J3 disc + E strap)	1
6	D8-05	Knee link with stop tooth	1
7	D8-06	Arm pad + pad mount + BOTTOM	1
8	D9-01	Drive bell crank	1
9	D9-02	Drive lobe ring (cam)	1
10	H-05	K hub cap with spring chamber	1
11	H-06	J2 cap (lobe ring seat)	1
12	H-01..04	Hub cap / spacers / ring nut (sheet HUB)	2+4+4+4
13	D8-07	Upper-arm cuff plate	1
14	MR117-2RS	Ball bearing 7 × 11 × 3 (hubs)	8
15	MR63ZZ	Ball bearing 3 × 6 × 2.5 (knee 4, follower 1)	5
16	683ZZ	Ball bearing 3 × 7 × 3 (crank pivot)	2
17	ISO 8734	Dowels: knee pins 3×10, stops 3×6 (3), J3 stop 2×5, nose 2.5×6	
18	K spring	Clock spring 0.3 × 5 × 145.6, 1.4310	1
19	misc	Pins, shims, e-clips DIN 6799 RS 2.3, wave washer, M4×6 ISO 10642 (	

Materials and finishes: on each part sheet.

NOTES

1. Right shoulder, rest pose. The 3-joint lead: verified score 93.35 (gap 18.28), held-out gap 22.38, complexity 18.5. J1, J2, J3 are the three counted (powered in MVP2) joints. K is slaved to J2 by the cam drive and is not counted; the two knee pins align the cuff and are not counted.
2. Cam drive: lobe ring D9-02 on the J2 cap pushes an MR63ZZ follower on the bell crank D9-01 (2 × 683ZZ on a Ø3 pivot in link 1b); the crank's Ø2.5 DLC nose drives the flank on link 1a. Below 7.28° of J2 the clock spring in the K cap (H-05, 0.08 N·m preload) holds K on stop A; above it K = 2.15 × (J2 − 7.28°), up to 50°.
3. Stops (all on a joint's own angle): J3 −111.90° to +122.74° (Ø2 dowel in a 250° groove under link 2's J3 ring); J2 −15.94° to +102.07° (Ø3 dowel in a 118° slot); K 0° / −62.85° (two Ø3 dowels); knee tooth at 154.78°.
4. Knee pins on MR63ZZ pairs (Ø6 J6 housings, Ø3 m6 pins in Ø3 H7 seats), already to the 2026-10-17 rule. J1 seats on your scapula handler at the acromion top (bracket on the J1 M18×0.5 spigot).
5. Bought parts (items 14-19) are listed, not drawn. Generated from the STEP files of the built CAD (gen 8) posed at its rest frame; status FOR REVIEW.

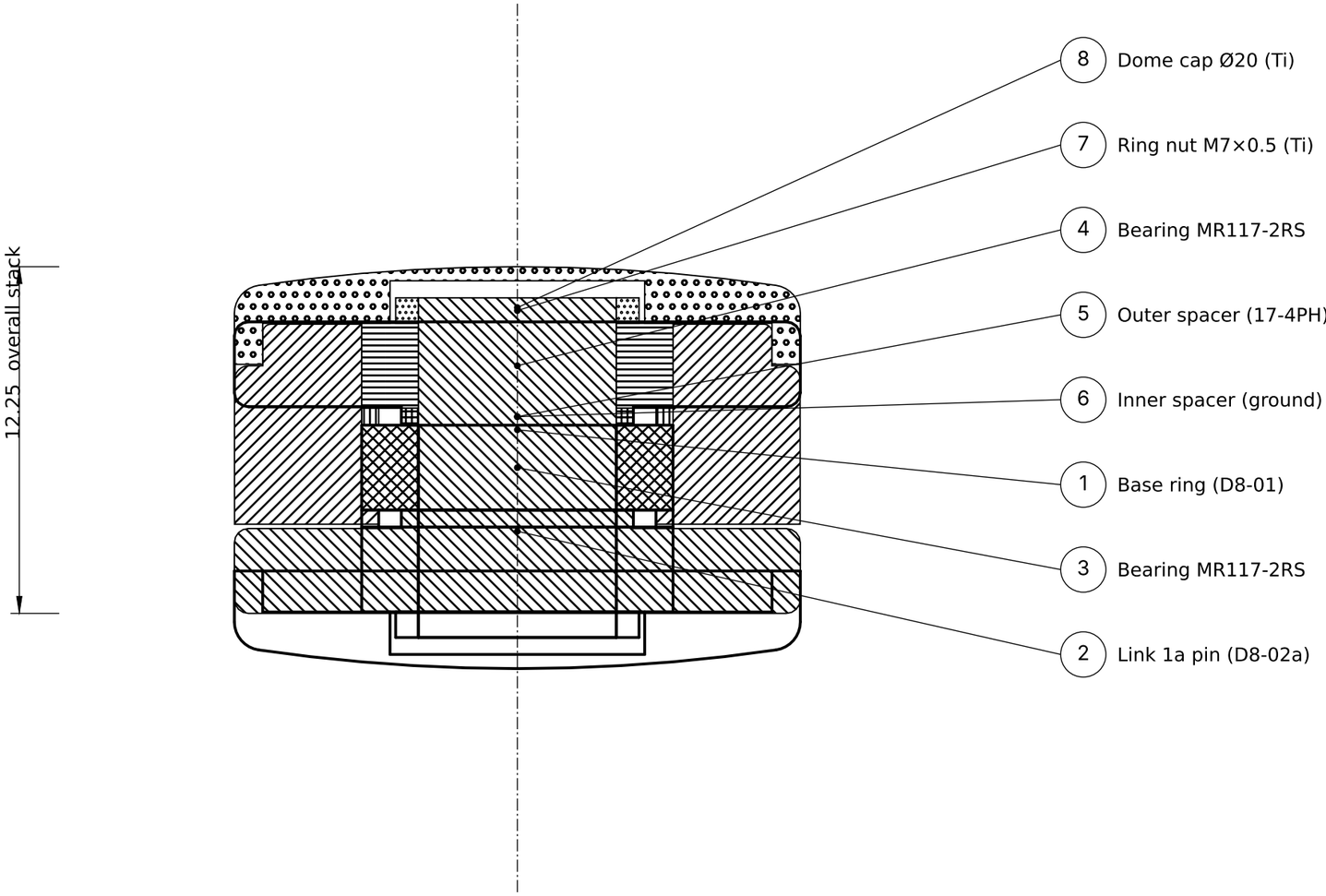
THIRD ANGLE

TITLE: **Cone-cut pin knee (3-joint lead), general arrangement**		DRAWING NO.: **EXO-A-GA**
DESIGN: conecut-knee-knuckle-3r (3-joint lead)	PART NO.: **A-ASSY**	REV / STATUS: A · for review
MATERIAL: see parts list	MASS: 216.1 g	QTY PER ASSEMBLY: 1
FINISH: see part drawings	SCALE: 1:2	SHEET: 2 / 15
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: pipeline/full-exo/designs/user_conecut

NOTES

1. Same stack at J1, K, J2 and J3 (K and J2 use their own caps H-05 / H-06): MR117-2RS pair back to back, shaft Ø7 j5 / housing Ø11 J6, preloaded through the ground inner spacer by the M7×0.5 ring nut.
2. 12.25 mm from the skin-side face to the crown (v5 standard, docs/HARDWARE.md).
3. At J1 your scapula handler's bracket threads onto M18×0.5 in place of the dome cap.
4. Drawn from the J1 hub of the built CAD; section plane through the axis, clipped 15 mm around it.

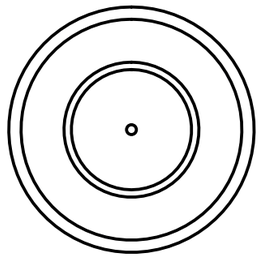
SECTION THROUGH THE HUB AXIS (5:1)



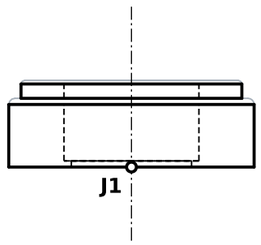
THIRD ANGLE

TITLE **Ø20 hub stack (v5), section**		DRAWING NO. **EXO-A-HUB**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **HUB-v5**	REV / STATUS A · for review
MATERIAL see part list	MASS see parts list	QTY PER ASSEMBLY 4
FINISH bearing seats unblasted	SCALE 5:1	SHEET 3 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE pipeline/full-exo/designs/user_conecut

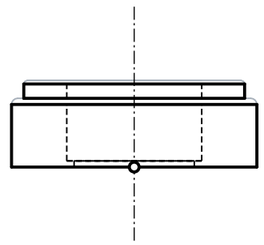
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)

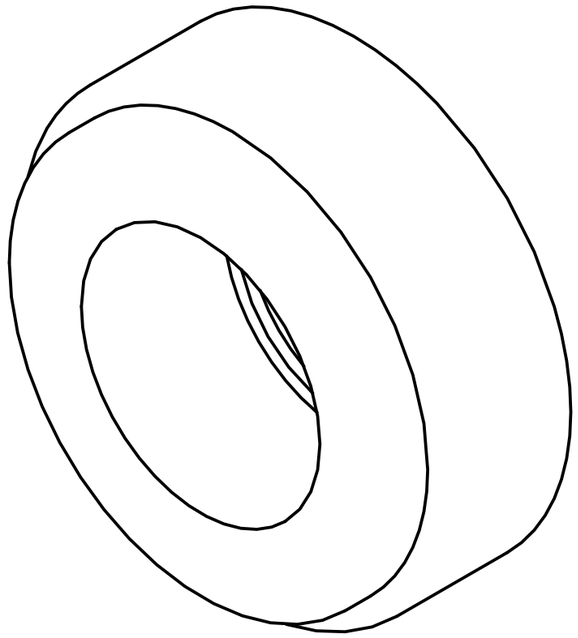


END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J1	0.00	0.00	0.00	0.00°	0.00°	-0.00°	0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

BUILT PART, ISOMETRIC (5:1), for reference



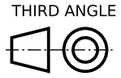
END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J1	Ø20	hub ring OD	pin / boss	0.00	5.10
J1	Ø9.80	hole	bore	0.00	0.50
J1	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
J1	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
J1	faces at s =	0.00, 0.50, 5.60, 7.07			

NOTES

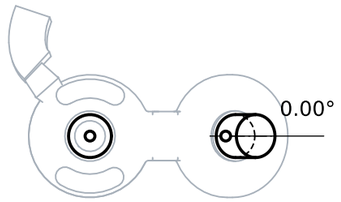
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 61_D8-01_scapula_base_AT_THE_J1_RING.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Your scapula handler's bracket threads onto the M18×0.5 spigot in place of the dome cap; no tab.
5. 5-axis machined.

THIRD ANGLE

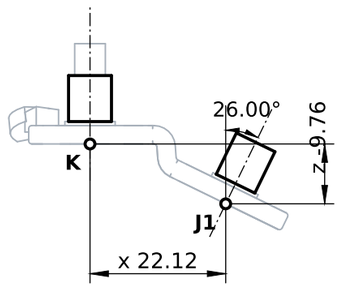


TITLE **Scapula base (J1 upper member)**		DRAWING NO. **EXO-A-D8-01**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **D8-01**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 6.50 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 4 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 61_D8-01_scapula_base_AT_THE_J1_RING

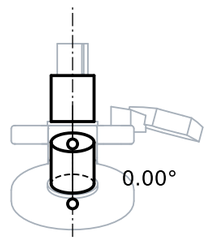
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
K	0.00	0.00	0.00	0.00°	0.00°	0.00°	-0.00°
J1	22.12	0.00	-9.76	26.00°	0.00°	26.00°	0.00°

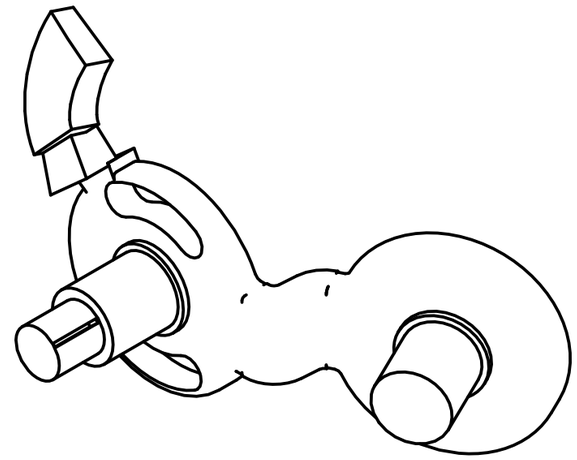
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
K–J1	26.00°	0.00	-55.12	-50.47

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (2:1), for reference



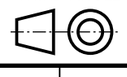
END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
K	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
K	faces at s =	0.00, 1.94, 3.00, 3.65, 11.15, 11.40, 16.30, 16.35			
J1	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
J1	faces at s =	0.00, 3.00, 3.65, 11.15			

NOTES

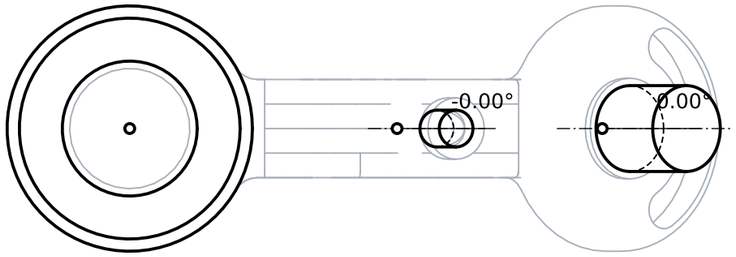
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 25_D8-02a_link_1a.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 5-axis machined from solid; R2 centre-line bends (tighter than the 15 mm bent-bar rule).
5. Flank lug: 4-axis UV wire EDM, profile ±0.02, a-C:H DLC 2-3 µm.

THIRD ANGLE

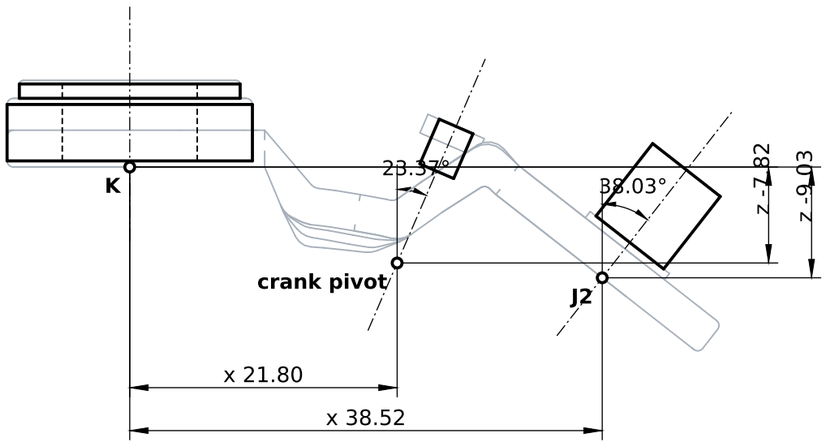


TITLE **Link 1a (J1 to K, flank lug)**		DRAWING NO. **EXO-A-D8-02a**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **D8-02a**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 12.55 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 1:1	SHEET 5 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 25_D8-02a_link_1a.step

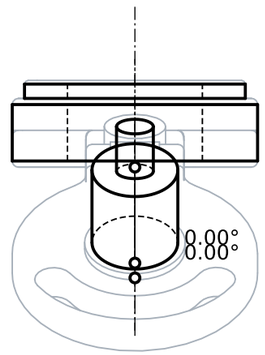
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
K	0.00	0.00	0.00	0.00°	0.00°	0.00°	0.00°
crank pivot	21.80	0.00	-7.82	23.37°	-0.00°	23.37°	-0.00°
J2	38.52	0.00	-9.03	38.03°	0.00°	38.03°	0.00°

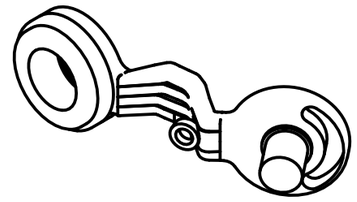
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
K–crank pivot	23.37°	0.00	-58.27	-54.95
K–J2	38.03°	0.00	-58.27	-62.52

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
K	Ø9.80	hole	bore	0.00	0.50
K	Ø20	hub ring OD	pin / boss	0.50	5.10
K	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
K	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
K	faces at s =	0.00, 0.50, 3.00, 3.50, 5.60, 7.07			
crank pivot	Ø3 H7	press, ISO 8734 m6 pin	bore	8.15	12.12
crank pivot	faces at s =	0.00, 2.99, 8.15, 10.47, 12.12			
J2	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
J2	faces at s =	0.00, 3.00, 3.65, 11.15			

CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
K	Ø3 H7	press, ISO 8734 m6 pin	7.75	12.6°	0.00 – 3.50
K	Ø3 H7	press, ISO 8734 m6 pin	7.75	-167.4°	-0.00 – 3.50

NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 26_D8-02b_link_1b.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Machined from solid (3+2 axis).
5. Crank pivot bore Ø3 H7 for the O3 h5/m6 17-4PH pin; J2 pin carries the 118° q2 stop slot.

THIRD ANGLE

TITLE **Link 1b (K to J2, crank pin)**		DRAWING NO. **EXO-A-D8-02b**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **D8-02b**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 13.93 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 6 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 26_D8-02b_link_1b.step

END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J2	0.00	0.00	0.00	0.00°	0.00°	0.00°	0.00°
J3	75.12	-0.00	-31.61	57.22°	5.45°	57.10°	8.39°

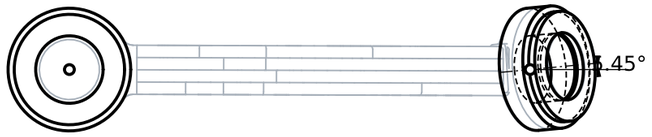
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

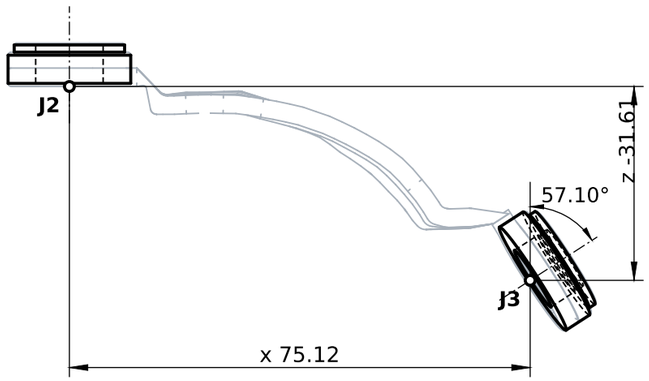
ENDS	ANGLE	AXIS GAP	t1	t2
J2–J3	57.22°	7.13	-79.77	-88.95

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

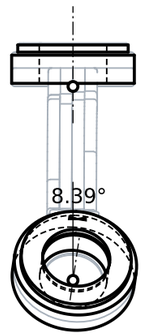
TOP (X right, Y up)



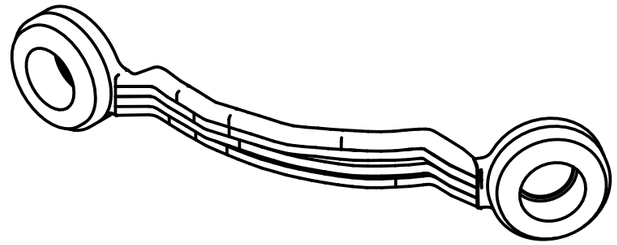
FRONT (X right, Z up)



RIGHT (Y right, Z up)



BUILT PART, ISOMETRIC (1:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J2	Ø9.80	hole	bore	0.00	0.50
J2	Ø20	hub ring OD	pin / boss	0.50	5.10
J2	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
J2	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
J2	faces at s =	0.00, 0.50, 3.00, 3.50, 5.60, 7.07			
J3	Ø9.80	hole	bore	0.00	0.50
J3	Ø20	hub ring OD	pin / boss	0.50	5.10
J3	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
J3	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
J3	faces at s =	0.00, 0.50, 1.00, 3.00, 5.60, 7.07			

CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
J2	Ø3 H7	press, ISO 8734 m6 pin	7.75	-51.7°	-0.00 – 3.50

NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 30_D8-03_link_2_(5-axis___3+2_machined_from.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 5-axis / 3+2 machined from solid; 68° drop at J2 (limit 70° on parts machined from solid).
5. J3 stop groove under the J3 ring: 2.10 +0.02/0 wide × 1.00 +0.05/0 deep, centre radius 7.500, 250° arc, full R1.05 ends; concentric to the J3 bore within 0.02.

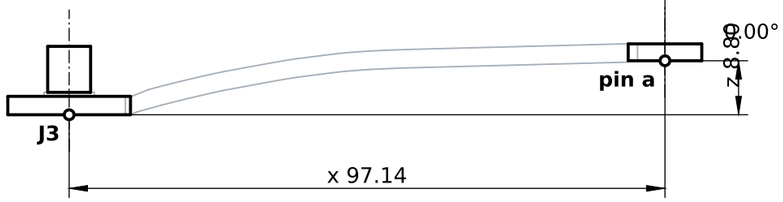
THIRD ANGLE

TITLE: **Link 2 (J2 to J3, J3 stop groove)**		DRAWING NO.: **EXO-A-D8-03**
DESIGN: conecut-knee-knuckle-3r (3-joint lead)	PART NO.: **D8-03**	REV / STATUS: A · for review
MATERIAL: Ti-6Al-4V (Grade 5)	MASS: 19.66 g	QTY PER ASSEMBLY: 1
FINISH: bead blast; mask bores, pins, threads	SCALE: 1:1	SHEET: 7 / 15
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: 30_D8-03_link_2_(5-axis___3+2_machine

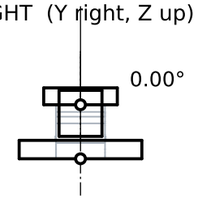
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J3	0.00	0.00	0.00	0.00°	0.00°	0.00°	-0.00°
pin a	97.14	0.00	8.80	0.00°	0.00°	0.00°	-0.00°

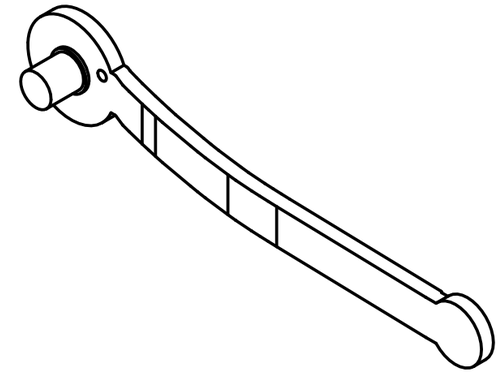
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
J3–pin a	parallel	97.14	–	–

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J3	Ø20	hub ring OD	pin / boss	0.00	3.00
J3	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
J3	faces at s =	0.00, 0.02, 3.00, 3.65, 11.15			
pin a	Ø12	boss	pin / boss	0.00	2.75
pin a	faces at s =	0.00, 2.75			

CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
J3	Ø2 −0.005/0	reamed, stop dowel press	7.50	18.4°	0.00 – 3.00

NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 45_D8-04_member_3.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. J3 stop dowel through hole Ø2.000 −0.005/0 reamed at r 7.5 (ISO 8734 2 m6 × 5, ground to 3.90).
5. E boss Ø12 × 2.75 with the Ø3 H7 seat of knee pin a.

THIRD ANGLE

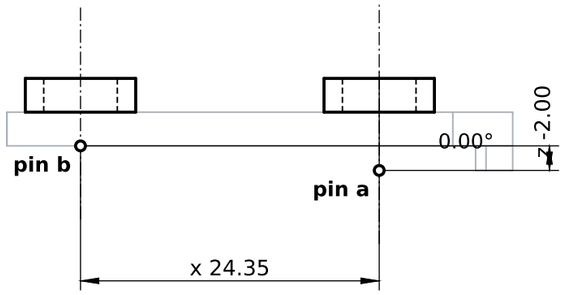


TITLE: **Member 3 (J3 disc + E strap)**		DRAWING NO.: **EXO-A-D8-04**
DESIGN: conecut-knee-knuckle-3r (3-joint lead)	PART NO.: **D8-04**	REV / STATUS: A · for review
MATERIAL: Ti-6Al-4V (Grade 5)	MASS: 15.71 g	QTY PER ASSEMBLY: 1
FINISH: bead blast; mask bores, pins, threads	SCALE: 1:1	SHEET: 8 / 15
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: 45_D8-04_member_3.step

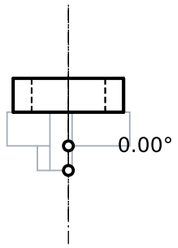
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
pin b	0.00	0.00	0.00	0.00°	0.00°	-0.00°	0.00°
pin a	24.35	-0.00	-2.00	0.00°	0.00°	-0.00°	0.00°

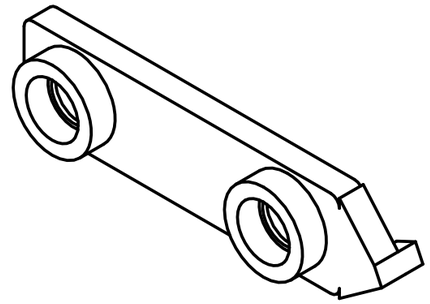
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
pin b–pin a	parallel	24.35	–	–

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (2:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
pin b	Ø6 J6	MR63ZZ housing	bore	0.00	5.50
pin b	Ø5.20	hole	bore	2.50	3.00
pin b	Ø9	bearing boss	pin / boss	2.75	5.50
pin b	faces at s =	2.50, 3.00, 5.50			
pin a	Ø6 J6	MR63ZZ housing	bore	2.00	7.50
pin a	Ø5.20	hole	bore	4.50	5.00
pin a	Ø9	bearing boss	pin / boss	4.75	7.50
pin a	faces at s =	0.00, 2.00, 4.50, 4.75, 5.00, 7.50			

NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 52_D8-05_knee_link.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Two Ø9 × 5.5 bosses, each 2 × MR63ZZ in Ø6 J6 housings with a 0.5 mm shoulder.
5. Stop tooth face ±0.02: meets the E strap at knee angle 154.78°.

THIRD ANGLE



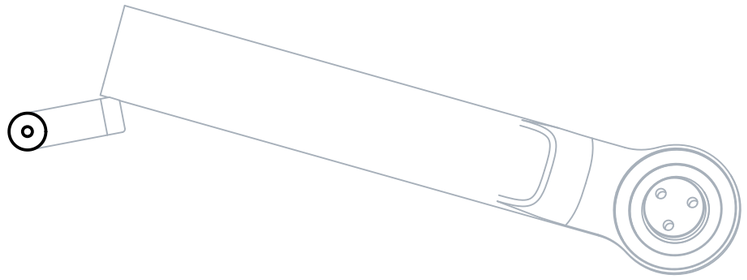
TITLE **Knee link (with stop tooth)**		DRAWING NO. **EXO-A-D8-05**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **D8-05**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 4.98 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 9 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 52_D8-05_knee_link.step

END POSITIONS (part frame)

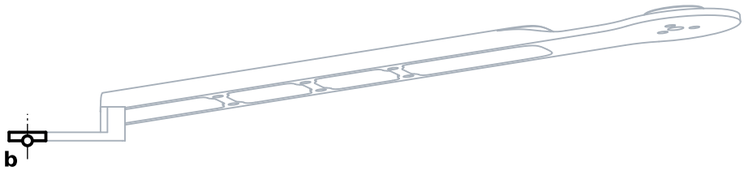
END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
pin b	0.00	0.00	0.00	0.00°	0.00°	-0.00°	-0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

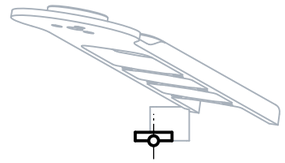
TOP (X right, Y up)



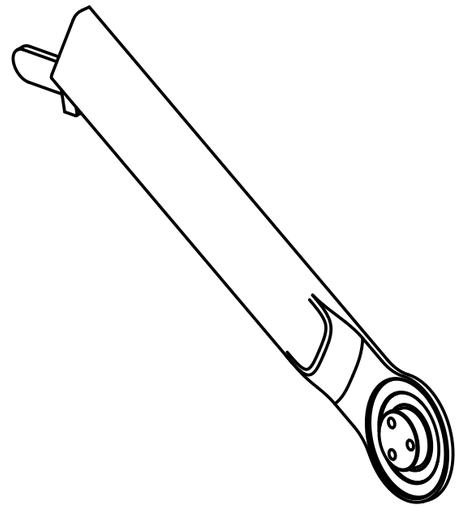
FRONT (X right, Z up)



RIGHT (Y right, Z up)



BUILT PART, ISOMETRIC (1:2), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
pin b	Ø12	boss	pin / boss	0.00	2.75
pin b	faces at s =	2.75			

CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 217.1, -23.6, 39.7		len 5.60
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 206.8, -20.7, 38.0		len 5.60
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 124.1, 8.3, 22.8		len 3.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 95.5, 16.3, 18.2		len 3.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 67.0, 24.4, 13.7		len 3.50
part	Ø5.10	hole	at 211.3, -25.4, 37.8		len 2.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 209.3, -31.0, 40.2		len 5.60
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 119.0, -8.3, 25.2		len 3.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 90.5, -0.3, 20.7		len 3.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 61.9, 7.8, 16.1		len 3.50

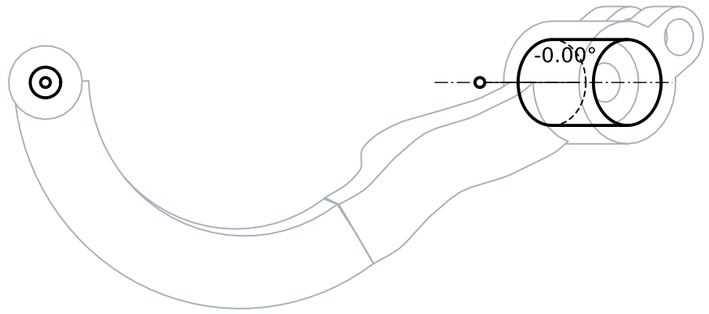
NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 54_D8-06_BOTTOM_+_arm_pad_+_pad_mount,_one_.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Your BOTTOM bar, cut square to the arm 19.8 mm distal of pin b and merged with the pad mount; the only one of your parts that is modified.
5. Ø3 H7 seat of knee pin b.

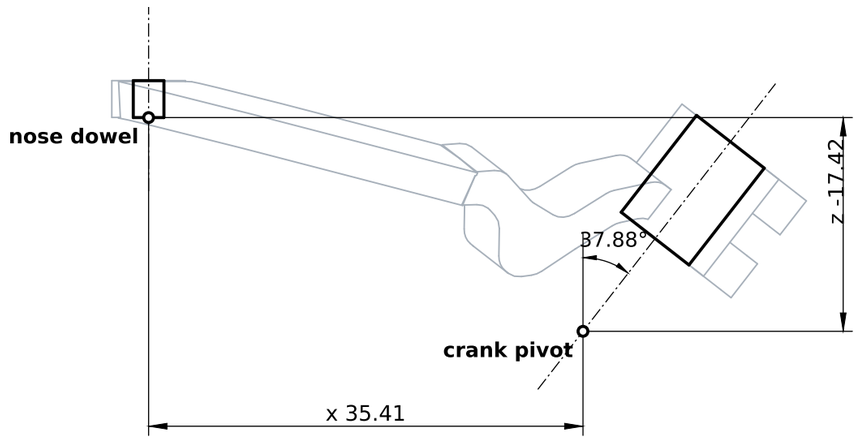
THIRD ANGLE

TITLE: **Arm pad + pad mount + BOTTOM (one part)**		DRAWING NO.: **EXO-A-D8-06**
DESIGN: conecut-knee-knuckle-3r (3-joint lead)	PART NO.: **D8-06**	REV / STATUS: A · for review
MATERIAL: 7075-T6	MASS: 58.66 g	QTY PER ASSEMBLY: 1
FINISH: clear anodise (never black)	SCALE: 1:2	SHEET: 10 / 15
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: 54_D8-06_BOTTOM_+_arm_pad_+_pad_m

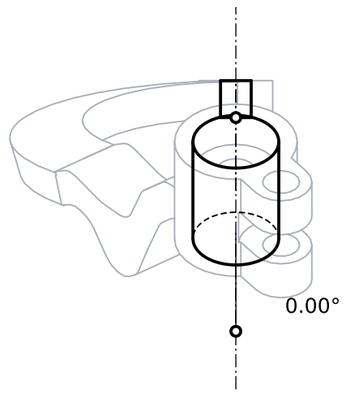
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
nose dowel	0.00	0.00	0.00	0.00°	0.00°	0.00°	0.00°
crank pivot	35.41	0.00	-17.42	37.88°	-0.00°	37.88°	-0.00°

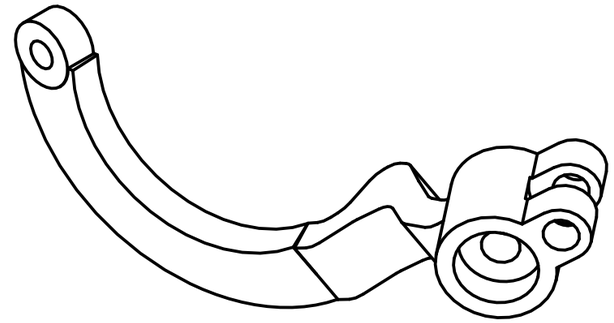
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
nose dowel–crank pivot	37.88°	0.00	-62.95	-57.68

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (2:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
nose dowel	Ø2.5 −0.005/0	reamed, nose dowel press	bore	0.00	3.01
nose dowel	faces at s =	0.01, 3.01			
crank pivot	Ø7 H7	bore	bore	9.57	19.57
crank pivot	Ø3.20	hole	bore	12.57	16.57
crank pivot	faces at s =	0.00, 9.57, 12.57, 13.07, 13.52, 16.07, 16.57, 19.57			

CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
crank pivot	Ø3 H7	press, ISO 8734 m6 pin	6.50	-145.4°	9.57 – 19.57

NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 32_D9-01_drive_bell_crank_(Ti-6Al-4V,_5-axi.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 5-axis machined from solid; all bores in one setup ±5 µm.
5. Straddle boss Ø10 × 10 with 683ZZ seats Ø7 J6 at both ends; Ø6 tip boss reamed Ø2.495-2.500 for the nose dowel (Loctite 648).

THIRD ANGLE

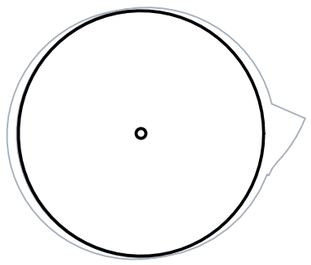
TITLE **Drive bell crank**		DRAWING NO. **EXO-A-D9-01**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **D9-01**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 7.38 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 11 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 32_D9-01_drive_bell_crank_(Ti-6Al-4V,_

END POSITIONS (part frame)

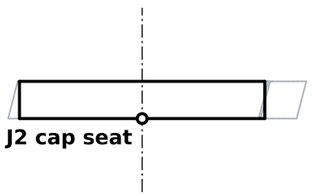
END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J2 cap seat	0.00	0.00	0.00	0.00°	0.00°	-0.00°	0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

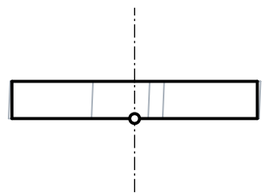
TOP (X right, Y up)



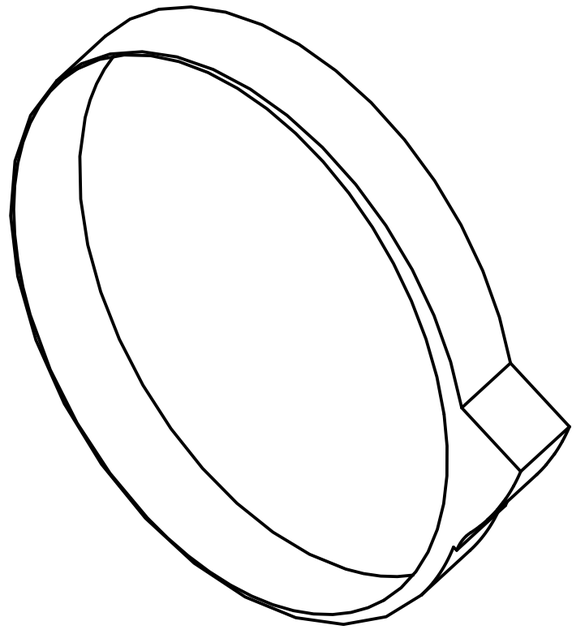
FRONT (X right, Z up)



RIGHT (Y right, Z up)



BUILT PART, ISOMETRIC (5:1), for reference



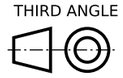
END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J2 cap seat	Ø20 H7	on the cap skirt	bore	0.00	3.03
J2 cap seat	faces at s =	0.00, 3.03			

NOTES

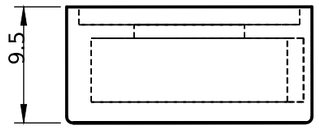
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 31_D9-02_drive_lobe_ring.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Wire EDM, taper 14.7° along the follower axis; profile r 10.65 → 12.62 (6 three-point arcs), dwell r 10.20.
5. Bore Ø20 H7 on the J2 cap skirt (Ø20 g6), keyed by a Ø1 dowel at the onset phase + 0.35° of q2.

THIRD ANGLE

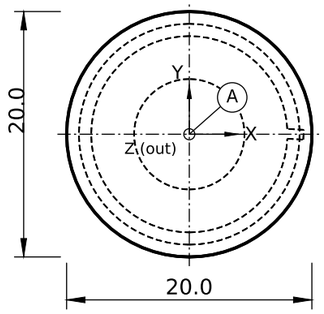


TITLE **Drive lobe ring (cam)**		DRAWING NO. **EXO-A-D9-02**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **D9-02**	REV / STATUS A · for review
MATERIAL 17-4PH H900	MASS 0.73 g	QTY PER ASSEMBLY 1
FINISH as wire-EDM, passivate	SCALE 2:1	SHEET 12 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 31_D9-02_drive_lobe_ring.step

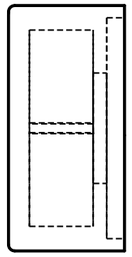
TOP

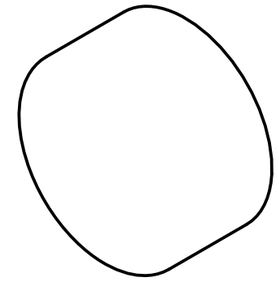


FRONT (looking at the outer face)



RIGHT





ISOMETRIC (2:1)

NOTES

1. Master geometry: STEP file 11_K_clock-spring_cap_O20_x_crown_F+18.25_(.step. Dimensions not shown: take from the model.
2. Bores and pins: the hole table. Fits from the design fit list (hub: Ø7 j5 shaft / Ø11 J6 housing for MR117-2RS pairs).
3. Position of tagged features: Ø0.02 to datum A for hub and pin features, Ø0.05 for others (proposed; review before release).
4. Surface: Ra 0.8 on bearing seats and pins, Ra 1.6 on mating faces, Ra 3.2 elsewhere. Break sharp edges 0.2 max.
5. Crown F+18.25 (6 mm taller than the v5 cap), spring chamber Ø16 × 5.2 with the outer-hook slot, M18×0.5.
6. Pinned to link 1b's ring at the spring's preload index.

HOLE AND PIN TABLE

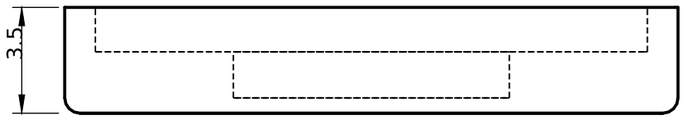
Origin: tag A axis, mid-length (datum A). X, Y, Z: part axes as drawn. Centres in mm.

TAG	SIZE / FIT	FUNCTION	TYPE	LEN	X	Y	Z	AXIS i, j, k
A1	Ø9.00	hole	bore	1.10	-0.00	-0.00	-2.47	-0.001 -0.000 -1.000
A2	Ø16.00	hole	bore	5.20	0.00	0.00	0.67	-0.001 -0.000 -1.000
A3	Ø18.00	hole	bore	1.45	-0.00	-0.00	-3.75	-0.001 -0.000 -1.000
A4	Ø20	hub ring OD	pin	8.95	0.00	0.00	0.00	+0.001 +0.000 +1.000

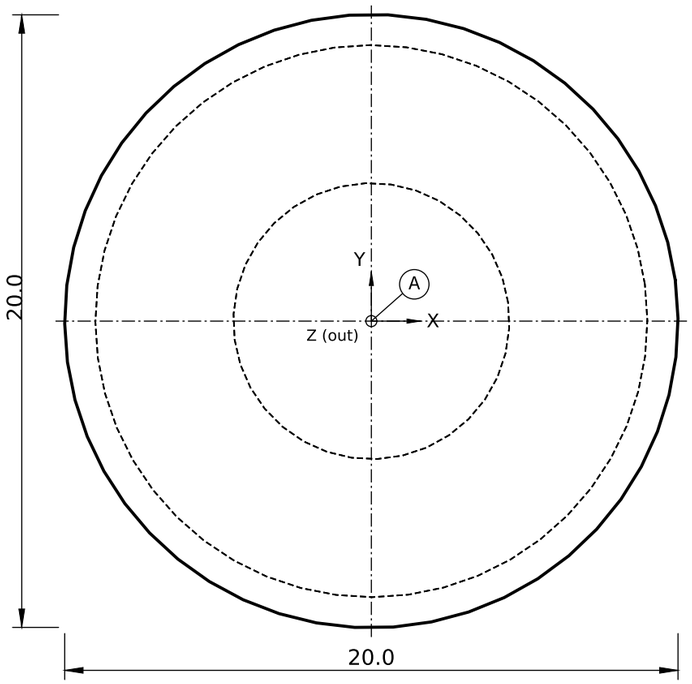
THIRD ANGLE

TITLE **K hub cap with spring chamber**		DRAWING NO. **EXO-A-H-05**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **H-05**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 6.54 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 13 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 11_K_clock-spring_cap_O20_x_crown_F+

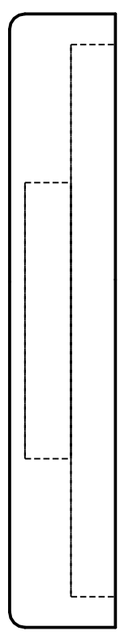
TOP

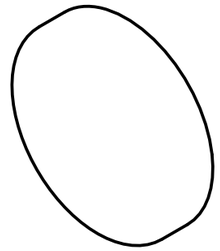


FRONT (looking at the outer face)



RIGHT





ISOMETRIC (2:1)

NOTES

1. Master geometry: STEP file 18_J2_cap_O20_x_crown_F+12.25,_flat_crown,_.step. Dimensions not shown: take from the model.
2. Bores and pins: the hole table. Fits from the design fit list (hub: Ø7 j5 shaft / Ø11 J6 housing for MR117-2RS pairs).
3. Position of tagged features: Ø0.02 to datum A for hub and pin features, Ø0.05 for others (proposed; review before release).
4. Surface: Ra 0.8 on bearing seats and pins, Ra 1.6 on mating faces, Ra 3.2 elsewhere. Break sharp edges 0.2 max.
5. Crown F+12.25, flat, R0.5 corner; skirt Ø20 g6 carries the lobe ring (D9-02). M18×0.5.

HOLE AND PIN TABLE

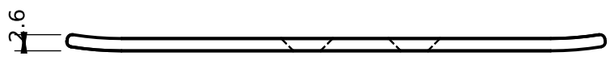
Origin: tag A axis, mid-length (datum A). X, Y, Z: part axes as drawn. Centres in mm.

TAG	SIZE / FIT	FUNCTION	TYPE	LEN	X	Y	Z	AXIS i, j, k
A1	Ø9.00	hole	bore	1.50	0.00	-0.00	0.73	+0.000 -0.000 -1.000
A2	Ø18.00	hole	bore	1.45	0.00	-0.00	-0.75	+0.000 -0.000 -1.000
A3	Ø20	hub ring OD	pin	2.95	0.00	0.00	0.00	-0.000 +0.000 +1.000

THIRD ANGLE

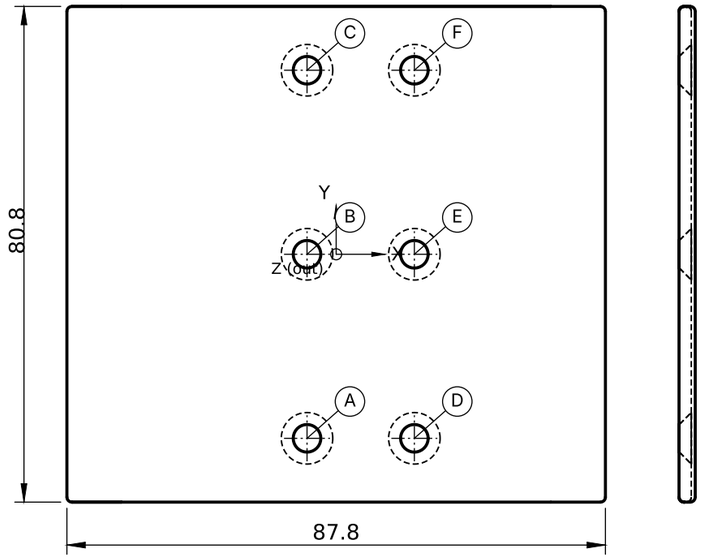
TITLE **J2 cap (lobe ring seat)**		DRAWING NO. **EXO-A-H-06**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **H-06**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 2.73 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 5:1	SHEET 14 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 18_J2_cap_O20_x_crown_F+12.25,_flat_c

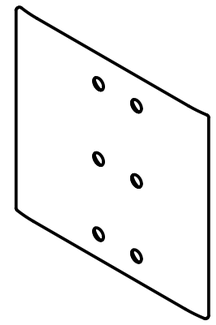
TOP



RIGHT

FRONT (looking at the outer face)





ISOMETRIC (1:2)

NOTES

1. Master geometry: STEP file 99_UPPER_ARM_CUFF_(one_sketch.step. Dimensions not shown: take from the model.
2. Bores and pins: the hole table. Fits from the design fit list (hub: Ø7 j5 shaft / Ø11 J6 housing for MR117-2RS pairs).
3. Position of tagged features: Ø0.02 to datum A for hub and pin features, Ø0.05 for others (proposed; review before release).
4. Surface: Ra 0.8 on bearing seats and pins, Ra 1.6 on mating faces, Ra 3.2 elsewhere. Break sharp edges 0.2 max.
5. Fixed to BOTTOM with 6 × M4×6 ISO 10642.

HOLE AND PIN TABLE

Origin: tag A axis, mid-length (datum A). X, Y, Z: part axes as drawn. Centres in mm.

TAG	SIZE / FIT	FUNCTION	TYPE	LEN	X	Y	Z	AXIS i, j, k
A	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	-4.75	-30.00	1.29	+0.000 +0.000 +1.000
B	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	-4.75	-0.00	1.29	+0.000 +0.000 +1.000
C	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	-4.75	30.00	1.29	+0.000 +0.000 +1.000
D	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	12.75	-30.00	1.29	+0.000 +0.000 +1.000
E	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	12.75	0.00	1.29	+0.000 +0.000 +1.000
F	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	12.75	30.00	1.29	+0.000 +0.000 +1.000

THIRD ANGLE

TITLE **Upper-arm cuff plate**		DRAWING NO. **EXO-A-D8-07**
DESIGN conecut-knee-knuckle-3r (3-joint lead)	PART NO. **D8-07**	REV / STATUS A · for review
MATERIAL AlSi10Mg (DMLS)	MASS 36.57 g	QTY PER ASSEMBLY 1
FINISH bead blast	SCALE 1:1	SHEET 15 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 99_UPPER_ARM_CUFF_(one_sketch.step